Planetary Mixer



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Planetary mixer

What it is used for

A planetary mixer mixes, whips, or kneads food ingredients using an attachment that rotates around its own axis while moving around the bowl. Use the correct attachment, speed, and batch capacity for the recipe and machine design.

Main parts

- Mixing bowl and bowl-locking mechanism
- Planetary mixing head and drive motor
- Beater, whisk, or dough hook attachments
- Speed selector and control panel
- Safety guard or bowl shield, if fitted
- Power switch and emergency stop, if fitted

Before you start

1. Confirm the mixer and attachment are suitable for the recipe and batch size.
2. Inspect the bowl, attachment, guard, locking mechanism, and power cable.
3. Ensure the bowl and attachments are clean and sanitised for food use.
4. Check that the machine is stable and all guards and safety devices work.
5. Verify the maximum capacity and recommended speed for the ingredients.
6. Keep utensils, loose clothing, and unnecessary items away from the mixing area.

How to use

1. Switch off the mixer before fitting or changing attachments.
2. Secure the bowl in its correct position and lock it as instructed.
3. Fit the appropriate beater, whisk, or dough hook securely.
4. Add the measured ingredients without exceeding the bowl capacity.
5. Lower the mixing head or close the guard according to the machine design.
6. Select the lowest suitable speed to begin mixing.
7. Increase speed gradually if the recipe and attachment permit it.
8. Observe the mixture and stop the machine if excessive vibration or unusual noise occurs.
9. Switch off and wait for the attachment to stop before scraping the bowl.
10. Use a suitable scraper only when the mixer is fully stopped and safe to access.
11. Switch off and isolate power before removing the bowl or attachment if required.
12. Transfer the mixture using clean utensils and follow food-safety requirements.
13. Clean and sanitise the bowl, attachments, and food-contact surfaces after use.

⚠ Safety

- Only trained personnel should operate commercial planetary mixers.
- Never place hands or utensils into the bowl while the attachment is moving.
- Do not bypass bowl guards, interlocks, or emergency stops.
- Switch off and isolate power before changing attachments or clearing jams.
- Do not exceed the specified capacity or use an attachment at an unsuitable speed.
- Keep the floor and controls dry and report damaged cables immediately.

Care & storage

- Clean and sanitise the bowl and attachments after each batch.
- Wipe the motor housing with an approved method; do not immerse it.
- Inspect the drive, attachment fittings, guards, and locking mechanism regularly.
- Lubricate service points only as specified by the manufacturer.
- Arrange servicing for abnormal noise, vibration, or speed-control problems.

Common mistakes to avoid

- Overloading the bowl or using the wrong attachment.
- Starting at excessive speed and causing splashing or ingredient overflow.
- Scraping or reaching into the bowl while the attachment is moving.
- Removing guards or changing attachments without isolating power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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